Chenru Wang

winterccc.github.io | winterccccc@gmail.com | Google Scholar

Education

Virginia Tech *Blacksburg, VA*

Doctor of Philosophy, Computer Science & Applications *Aug. 2026 – Present (Expected Dec. 2029)*

Ocean University of China *Qingdao, China*

Bachelor of Science, Mathematics and Applied Mathematics *Sept. 2021 – Jun. 2025*

Research Interests

Generative AI, Reinforcement Learning, and Trustworthy AI

Publications

[1] **Wang, C.**, Chen, Y., Yang, Z., Zhou, J. T., Zhang, C. (2026). IMS3: Breaking Distributional Aggregation in Diffusion-Based Dataset Distillation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**). **[PDF] [Code]**

[2] **Wang, C.**, Zhu, B., Zhang, C. (2026). Free Lunch for Stabilizing Rectified Flow Inversion. *International Conference on Learning Representations* (**ICLR**). **[PDF] [Code]**

[3] Jiang, W., Cai, Z., Shao, Y., **Wang, C.**, Han, B., Song, Z., Chen, K., An, S., Yang, X., Yang, Z. (2026). ManimAgent: Self-Evolving Multimodal Agents for Visual Education. *Conference on Empirical Methods in Natural Language Processing* (**EMNLP**). **[PDF] [Homepage] [Code]**

[4] **Wang, C.**, Chen, Z., Li, M., Yin, H., Zhou, S., Zhang, J., Zeng, X., Zhang, Q. (2025). DDUM: Deformable Dilated U-structure Module for Coronary Stenosis Detection. *Medical Engineering & Physics*. **[PDF]**

[5] Jiang, W.*, **Wang, C.***, Wang, Y., Yang, Y., Han, B., Chen, K., An, S., Yang, Z., Zhang, C. ScoutSampler: Learning to Find Causally Critical Frames for Long Video Understanding. (2026). *NeurIPS 2026 Workshop on Grounded and Faithful Vision-Language Models for Real-World Deployment.* (**NeurIPS Workshop**)

Preprints & Under Review

[6] **Wang, C.**, Han, Y., Zhang, X., Yu, G., An, S. X-OPD: Rethinking Multi-Teacher On-Policy Distillation in Diffusion Models. (ICLR 2027 Submission)**[Homepage]**

[7] **Wang, C.**, Samptur, N., Chou, S., Zhai, S., Chen, P., Zhang, X., An, S. Rethinking Backdoors in Text-to-Image Generation Models: When, Why, and How. (NDSS 2027 Submission)

[8] **Wang, C.**, Jiang, W., Li, A., Han, B., Wei, X., Chen, K., Yang, Z., An, S. To Ask or Not to Ask: Backdooring When a Cheap Agent Consults Its Expert. (ICLR 2027 Submission)

[9] Chen, Y.*, **Wang, C.***, Ye, X., Zhou, T., Zhang, C. MGPO: Dual-Space Manifold Alignment for Diffusion-Based Dataset Distillation via Policy Optimization. (ICLR 2027 Submission)

[10] Li, A.*, **Wang, C.***, An, S. Immunized Against a Simulated Attacker: Model-Blind Noise Weakens Release-Time Model Immunization. (ICLR 2027 Submission)

[11] Zhang, H.*, **Wang, C.***, Ye, X., Wei, X., Zhang, C. Gap-Targeted Synthesis: Guiding Diffusion Models to Fill Distributional Gaps in Real Training Data. (ICLR 2027 Submission)

* Equal contribution.

Experience

Research Assistant — *Blacksburg, VA*

Virginia Tech [Advisor: Shengwei An] — *Nov. 2025 – Present*

- Introduced a new threat model in which backdoors implanted in an upstream diffusion model propagate to downstream classification and vision-language models, exposing a previously overlooked attack surface in generative-model supply chains (NDSS 2027 Submission).
- Demonstrated a backdoor attack on a cost-tiered SWE-agent system in which a cheap agent decides when to consult a stronger expert model, turning the consultation policy itself into an exploitable attack surface (ICLR 2027 Submission).

Research Intern — *Beijing, China*

AIGC Foundation Model Dept., StepFun [Advisor: Gang Yu] — *Mar. 2026 – Aug. 2026*

- **Post-trained an 8B proprietary image generation foundation model using on-policy distillation and reinforcement learning on distributed clusters of up to 128 GPUs, improving sampling efficiency and human-preference alignment.**
- Designed X-OPD, a multi-teacher on-policy distillation framework for diffusion models, reducing sampling to 4 steps while retaining teacher quality (ICLR 2027 Submission).
- Built large-scale data curation and cleaning pipelines to improve the quality of foundation model training data.

Research Assistant — *Hangzhou, China*

AGI Lab, Westlake University [Advisor: Chi Zhang] — *Apr. 2025 – Mar. 2026*

- Developed an inversion-based fine-tuning method that improves the quality and diversity of distilled datasets (IMS3, **CVPR 2026**).
- Proposed the first reinforcement learning framework for dataset distillation, unifying diverse downstream objectives into a single policy-optimization pipeline (MGPO).
- Developed two plug-and-play velocity correction methods that improve inversion stability and editing fidelity in flow-based generative models, achieving state-of-the-art results on PIE-Bench (**ICLR 2026**).
- Developed an RL-based post-training method for long video understanding, improving long-range temporal reasoning (ScoutSampler, NeurIPS 2026 Workshop).

Undergraduate Research Assistant — *Qingdao, China*

Ocean University of China [Advisor: Xueying Zeng] — *Nov. 2022 – Apr. 2025*

- Proposed the Deformable Dilated U-structure Module to adapt generic networks for coronary artery stenosis detection and improve generalization (DDUM, ***Medical Engineering & Physics*** 2025).
- Constructed a more complex and representative clinical dataset (302 dataset) to evaluate model robustness in real-world scenarios.

Competitions & Awards

Bronze Medal, Nemotron Model Reasoning Challenge (Kaggle, hosted by NVIDIA), 2026

Skills

Programming: Python (PyTorch, Diffusers, Accelerate), C/C++

ML & Research Tools: HuggingFace, Weights & Biases (wandb), OpenCV, scikit-learn

AI Research Workflow: Designed and maintained a standardized research workflow integrating frontier coding agents and LLMs across ideation, implementation, experimentation, and analysis.

Service

Conference Reviewer: AAAI 2026, AAAI 2027, CVPR 2026, ICLR 2027, ICML 2026 (Golden Reviewer), NeurIPS 2026